TUMMALA YASWANTH SAI

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Right to work in the UK: Graduate Visa (valid until Sep 2028)
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PROFESSIONAL SUMMARY

MSc Data Science graduate with Distinction (Teesside University, 2026) and a B.Tech in Information Technology. Developed a fraud detection pipeline on 5 million synthetic transactions (Random Forest, recall 0.69, ROC-AUC 0.93) and deployed it as a Flask web app, and a CNN-LSTM image captioning model in TensorFlow. Proficient in Python (pandas, NumPy, scikit-learn) and Power BI. Certified in AWS Machine Learning Foundations, Google Cloud Digital Leader and Azure AZ-900.

CORE SKILLS

Programming	Python (pandas, NumPy, scikit-learn, XGBoost, imbalanced-learn), TensorFlow, Flask, Plotly
AI / Machine Learning	Supervised learning, classification, deep learning, neural networks (CNN, LSTM), computer vision, NLP, transfer learning, feature engineering, SMOTE, threshold tuning, model evaluation (precision, recall, F1, ROC-AUC, BLEU), model deployment
Data & Analytics	SQL, data preprocessing and cleaning, Power BI (Power Query, DAX), KPI dashboards
Cloud & Tools	AWS, Google Cloud, Microsoft Azure (foundation certified), Jira, Microsoft Excel, PowerPoint

CERTIFICATIONS

AWS Academy Graduate, Machine Learning Foundations • AWS Academy Graduate, Cloud Foundations
Google Cloud Certified, Cloud Digital Leader • Microsoft Azure Fundamentals (AZ-900)

PROJECTS

Fraudulent Financial Transaction Detection (MSc Dissertation)

- Developed an end-to-end machine learning pipeline in Python (pandas, NumPy, scikit-learn) on a 5-million-row synthetic transaction dataset with a 3.59% fraud rate.
- Engineered 11 behavioural features (8 of the top 10 model features) and applied SMOTE to training data only, raising Random Forest recall from below 0.20 to 0.69.
- Compared six classifiers and tuned decision thresholds over 91 values, selecting Random Forest as best (F1 0.51, recall 0.69, ROC-AUC 0.93).
- Designed a Flask web app with six Plotly dashboards and a live fraud-prediction page for non-technical users.

Transforming Remote Sensing Images into Textual Descriptions (B.Tech Major Project, team of three)

- Developed a deep learning image captioning model in Python and TensorFlow, combining a pre-trained Inception V3 CNN (transfer learning) with an LSTM to generate text descriptions of images.
- Trained the model on Google Colab for 50 and 100 epochs, raising BLEU-1 from 0.34 to 0.36 and BLEU-2 from 0.16 to 0.19.
- Implemented gTTS text-to-speech to read generated captions aloud, making image content accessible to visually impaired users.

EXPERIENCE

Data Visualisation Intern, MSc Advanced Practice Internship

Teesside University | Sep 2025 - Jan 2026

- Developed 6 interactive Power BI dashboards as part of a team project on machine learning and deep learning health status prediction, visualising workout, diet and health data and model outputs.
- Leveraged SQL and Power Query to prepare data, then built the data model and wrote DAX measures for the KPIs.
- Collaborated with the ML/DL team to reconcile dashboard metrics with model outputs, and documented each dashboard.

AI-ML Virtual Intern

AICTE & EduSkills, supported by AWS Academy | May 2023 - Jul 2023 (10 weeks)

- Passed module exams across a 10-week AI-ML programme covering machine learning algorithms, data preprocessing and model evaluation.
- Executed hands-on tasks from data preparation to model training and testing, leveraging Python ML libraries and AWS cloud-based ML tools.

EDUCATION

MSc Data Science with Advanced Practice (Distinction) | Teesside University, Middlesbrough | 2024 - 2026

B.Tech Information Technology | Velagapudi Ramakrishna Siddhartha Engineering College, Vijayawada | 2020 - 2024